بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

LIVE ONLINE COURSE | COHORT FALL 2026

AI & Machine Learning for Civil Engineers

SESSION 01 · PYTHON AS A TOOL, NOT A BARRIER

From Engineering Problems to Data & Machine Learning

Python & Google Colab: a practical, no-installation starting point for engineering data and machine learning.



Dr. Abdul Waris
Research Associate · Universität Siegen, Germany

Technology partner QUDELTA STUDIOS

WELCOME TO THE COHORT

Machine learning, taught the way engineers think.

8
live, instructor-led sessions

90 min
each, about half of it coding together

4 weeks
from first cell to uncertainty analysis

8
hands-on Colab notebooks, one per session

1
real engineering dataset in every session

IN THE ROOM

B.Tech students | M.Tech students | Ph.D. scholars | Faculty | Practising engineers | **No coding experience needed**



YOUR INSTRUCTOR

Dr. Abdul Waris

Research Associate · Universität Siegen, Germany

Ph.D. in Geotechnical Engineering
Indian Institute of Technology (IIT) Hyderabad

Former Prime Minister’s Research Fellow
PMRF, Government of India

Research in explainable ML & reliability
The basis for sessions 07 and 08 of this course

Invited talks
Oregon State University · DY Patil · NIT Srinagar

THE COURSE AT A GLANCE

Eight sessions. One engineering workflow.

PART 1 · FOUNDATIONS

01 **From Engineering Problems to Python** TODAY
AI/ML basics, Google Colab, NumPy & pandas

02 **From Raw Engineering Data to an ML-Ready Dataset**
EDA, data quality, outliers, train/test split, leakage

03 **Your First Engineering Prediction Model**
Linear regression, error metrics, cross-validation

04 **Nonlinear Engineering Problems: Trees & Ensembles**
Decision trees, Random Forest, boosting

PART 2 · PREDICTION TO DECISION

05 **From Prediction to Engineering Decision: Classification**
Logistic regression, confusion matrix, precision & recall

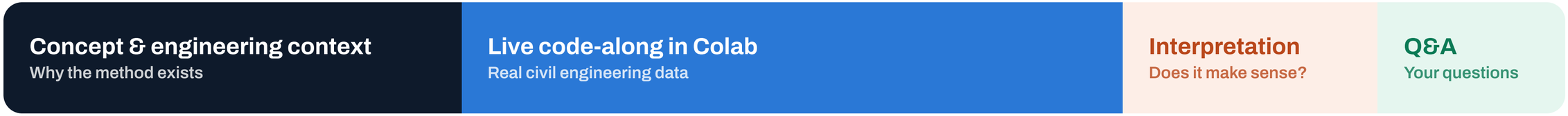
06 **Artificial Neural Networks**
Architecture, activations, training, overfitting

07 **Can We Trust Our ML Model?**
Feature importance, SHAP, engineering judgement
Instructor's research area

08 **From Prediction to Probability**
Monte Carlo, ML surrogates, probability of failure
Instructor's research area

HOW EVERY SESSION RUNS

Concept, then code, then judgement.



One notebook per session

Structured Colab notebooks you keep and revisit.

Real engineering data

Materials, structures, buildings & energy, transport, water & environment, geotechnics.

Recordings

Shared after each session; available until two weeks after the course ends.

Engineering judgement

Validation, limitations and physical plausibility, not just accuracy.

TODAY · SESSION 01

From engineering problems to Python

01 **From engineering problems to machine learning**
AI, ML and DL · supervised and unsupervised learning · one workflow



02 **Python & Google Colab**
Why Python · notebooks · Colab vs alternatives · our toolkit





03 **Hands-on: our first cells**
Open the notebook · run code · load data · first plot





BY THE END OF TODAY YOU CAN

- Place AI, ML and deep learning on one map
- Tell regression, classification and unsupervised problems apart
- Describe the ML workflow for a civil engineering problem
- Open, run and save a Colab notebook

01

PART 01

From engineering problems to machine learning

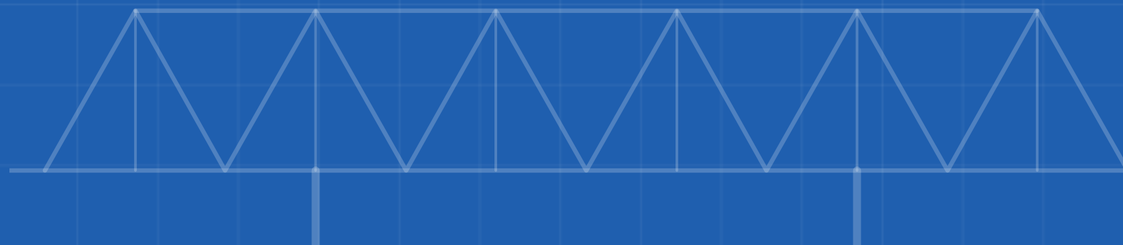
a A problem we already know

b AI, ML and deep learning

c Supervised & unsupervised

d From training to prediction

e One ML workflow



START WITH AN ENGINEERING PROBLEM

Can we predict concrete strength from its mix and age?

X inputs: mix and age

Cement
102–540 kg/m³

Water
122–247 kg/m³

Fly ash
0–200 kg/m³

Slag
0–359 kg/m³

Aggregates
coarse 801–1145 · fine 594–993 kg/m³

Superplasticizer
0–32 kg/m³

Age
1–365 days

WE MEASURE A RESPONSE

y **Compressive strength** 2–83 MPa

We already understand this problem. We have the ingredients, we have age, and we measure a response.

THE QUESTION IS

Can we use historical examples to learn the relationship?

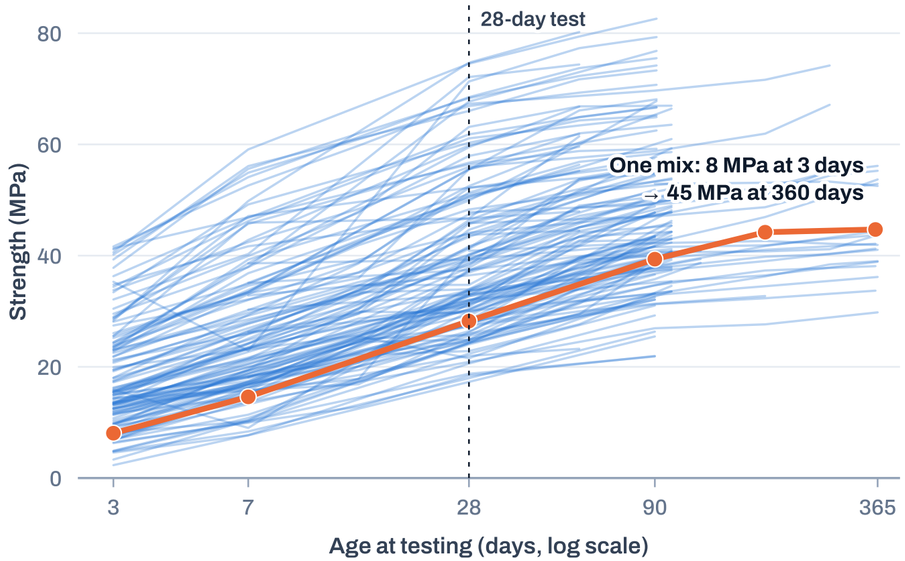
Ranges from the Concrete Compressive Strength dataset, UCI Machine Learning Repository (Yeh, 1998): 1,030 laboratory mixes.

WHAT 1,030 LABORATORY TESTS SHOW

Each input tells part of the story

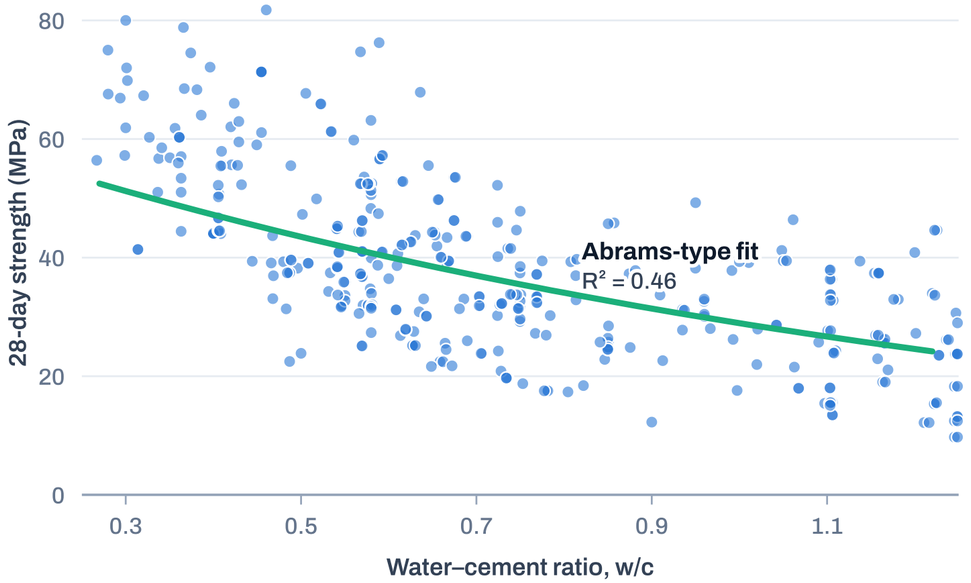
Strength keeps growing with age

132 mixes tested at four or more ages. Highlighted: one mix went from 8 MPa at 3 days to 45 MPa at 360 days.



Lower water–cement ratio, higher strength

An Abrams-type curve on 425 mixes at 28 days explains only **46%** of the variation.

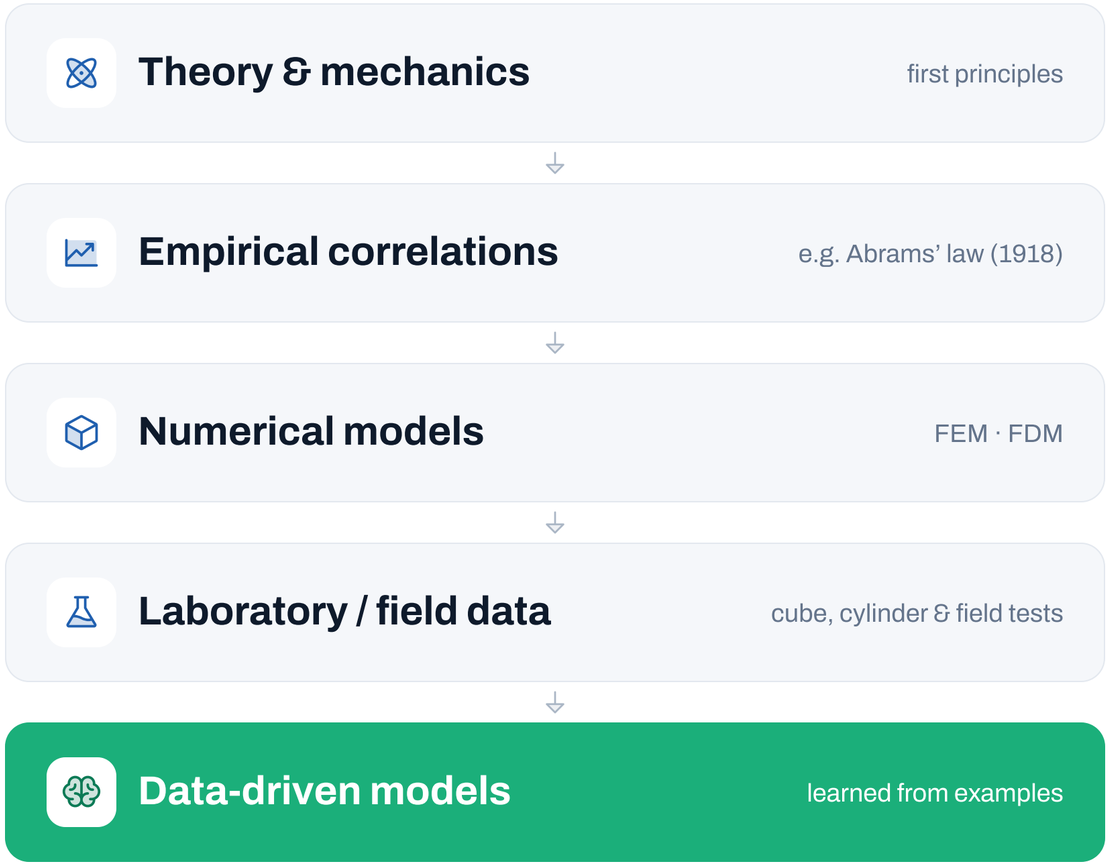


One variable at a time leaves most of the story untold. Machine learning can use all the inputs together.

Data: UCI Concrete Compressive Strength (Yeh, 1998). Curve: $f_c = A / B^{w/c}$ fitted to 28-day mixes, A = 65.4 MPa, B = 2.26.

CIVIL ENGINEERS ALREADY BUILD MODELS

ML is another engineering modelling tool.



Machine learning learns relationships from examples in data.

We already build models. We use equations, correlations, numerical models, experiments and design standards.

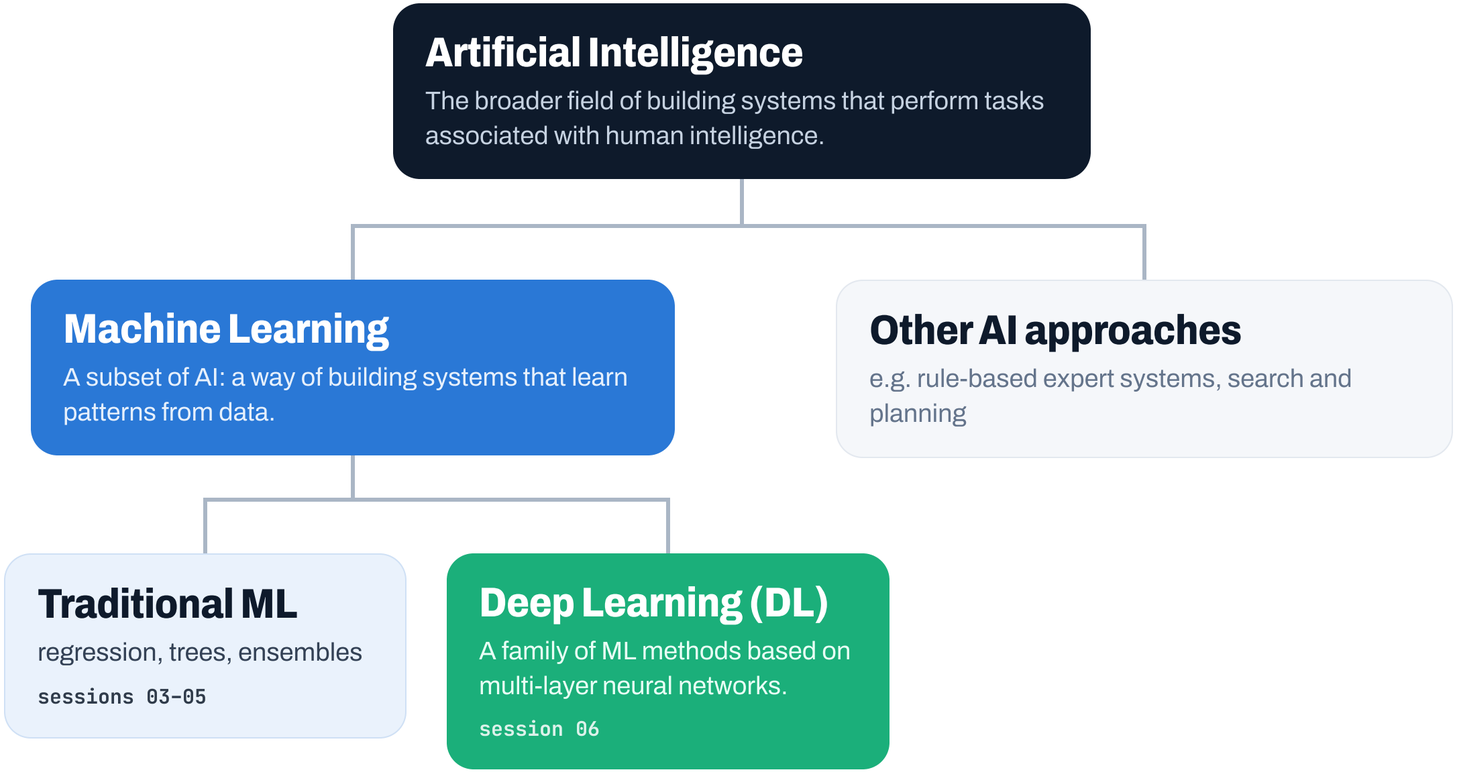
Machine learning doesn't suddenly replace engineering thinking. It gives us another way to represent relationships **when we have sufficient data.**

MACHINE LEARNING IS ANOTHER MODELLING APPROACH

The important difference is that instead of prescribing the relationship ourselves, we allow an algorithm to learn a relationship from examples in data.

THE VOCABULARY · YOU WILL HEAR THESE TERMS CONSTANTLY

Artificial intelligence is the broad idea.



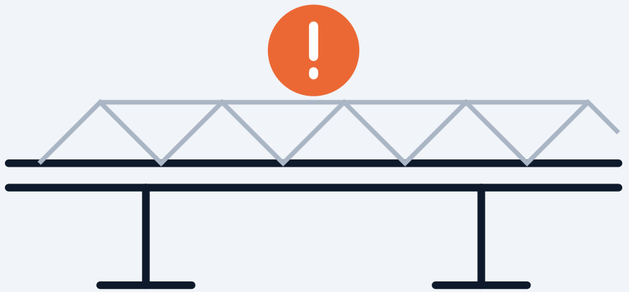
OUR FOCUS

Machine learning is the focus of this course. We meet deep learning with artificial neural networks.

1956	“Artificial intelligence” becomes a field at the Dartmouth summer workshop
1959	Arthur Samuel popularises the term “machine learning”
2012	Deep neural networks break through in image recognition

ONE ENGINEERING EXAMPLE · STRUCTURAL INSPECTION

Same engineering world. Different levels.



LEVEL 1 · ARTIFICIAL INTELLIGENCE (AI)

Can a system assist an engineer in deciding whether a structure requires attention?

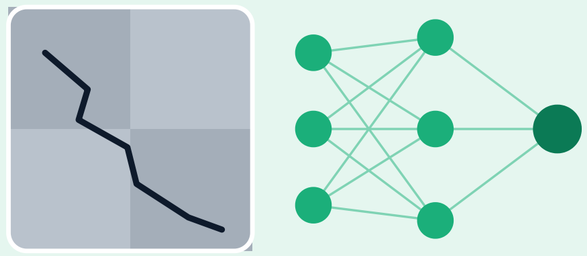
AI = the broader goal



LEVEL 2 · MACHINE LEARNING (ML)

Can we learn from historical inspection data to classify defects?

ML = learning from data



LEVEL 3 · DEEP LEARNING (DL)

Can a neural network learn complex patterns directly from images of structural defects?

DL = neural-network-based ML

MACHINE LEARNING, DEFINED

> **Field of study that gives computers the ability to learn without being explicitly programmed.**

Arthur Samuel | 1959 | IBM pioneer of the self-learning checkers program

MACHINE LEARNING ALGORITHMS

Two families we will use

SUPERVISED LEARNING

Learns from the “right answers”

Every example comes with its label ***y***.

X_1	X_2	X_3	y · label

Regression · Classification

UNSUPERVISED LEARNING

Finds structure on its own

Data only comes from inputs ***X***, with no output label.

X_1	X_2	X_3	no label

Clustering · Anomaly detection · Dimensionality reduction

SUPERVISED LEARNING · EXAMPLES

Input X → output Y

Input (X)		Output (Y)	APPLICATION
Email	→	`Spam? (0/1)`	Spam filtering
Audio	→	`Text transcripts`	Speech recognition
English	→	`Arabic, Urdu, Spanish, etc.`	Machine translation
Ad, user info	→	`Click? (0/1)`	Online advertising
Mix design + age	→	`Strength (MPa)`	**Concrete mix design**
Photo of a concrete surface	→	`Crack? (0/1)`	**Visual inspection**

DeepLearning.AI. Phoon, K. K., Ching, J. & Cao, Z. (2022). Unpacking data-centric geotechnics. *Underground Space* 7(6), 967–989. doi:10.1016/j.undsp.2022.04.001. Highlighted rows: civil engineering examples added for this course.

SUPERVISED LEARNING · LEARNS FROM BEING GIVEN THE “RIGHT ANSWERS”

Regression or classification?

34.7 MPa

REGRESSION PROBLEM

Predict a number

from infinitely many possible outputs.

CIVIL ENGINEERING EXAMPLES

- Concrete strength
- Settlement
- Traffic volume
- Energy demand
- Factor of safety

Stable Unstable

CLASSIFICATION PROBLEM

Predict a category

from a small number of possible outcomes.

CIVIL ENGINEERING EXAMPLES

- Stable / unstable
- Defect type
- Soil class
- Damage category

The target determines the ML problem.

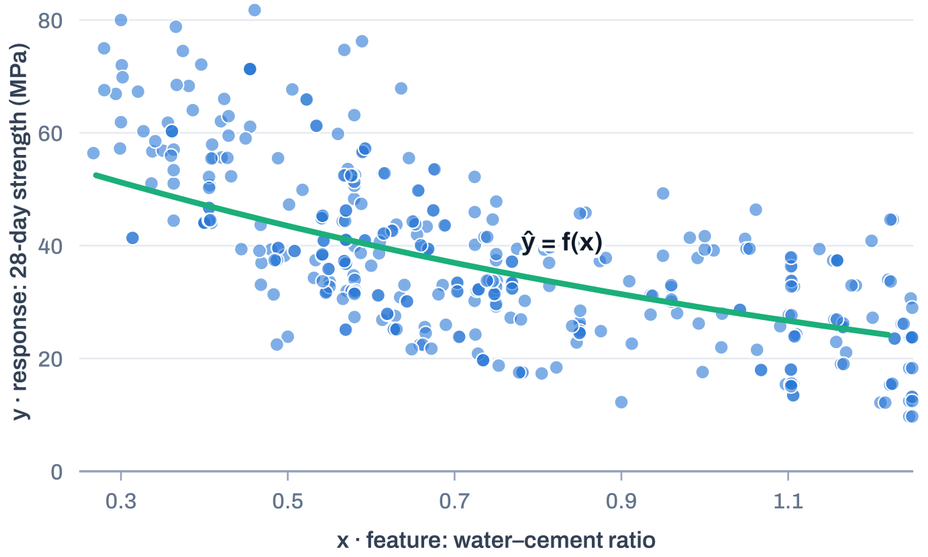
SUPERVISED LEARNING · WHAT THE MODEL LEARNS

Fit a curve, or draw a boundary

REGRESSION PROBLEM

Predict the strength itself

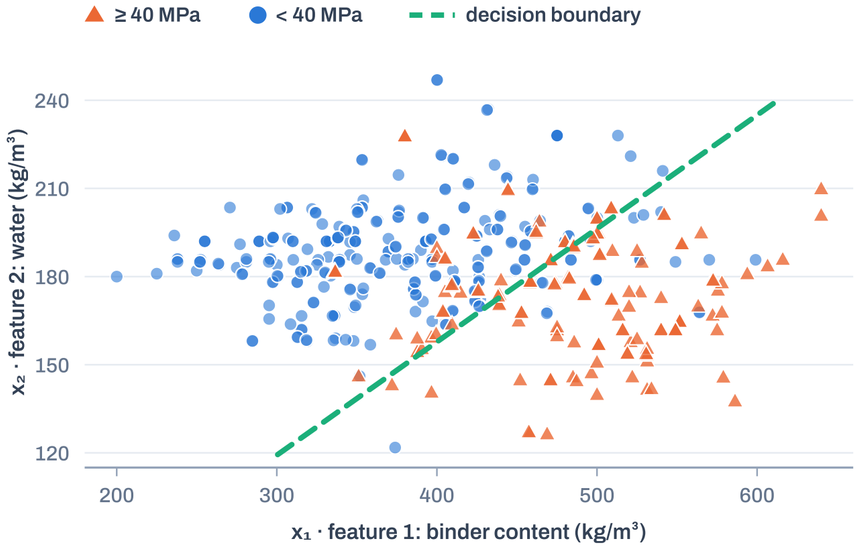
One feature ***x***, response variable ***y***, model output $\hat{y}$.



CLASSIFICATION PROBLEM

Will the mix reach 40 MPa?

Two features x_1, x_2. A simple model is right 84% of the time (always guessing "below": 66%).



Same 425 mixes at 28 days. Change the target, and the problem changes.

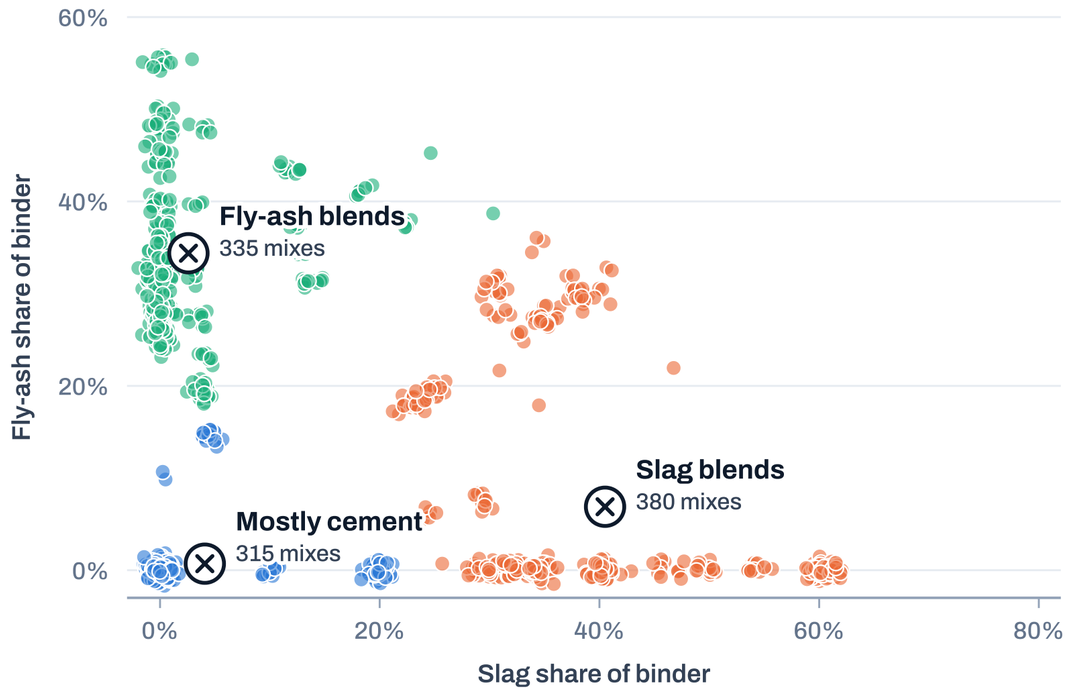
Data: UCI Concrete Compressive Strength (Yeh, 1998). Binder = cement + slag + fly ash. Boundary from logistic regression. Concept after DeepLearning.AI.

UNSUPERVISED LEARNING

Takes data without labels and tries to automatically group them into clusters.

No strength labels. Three binder families emerge.

k-means on 1,030 mixes, using only how the binder is made up.



Data only comes from **Input (X)** but not output label (Y). Algorithm has to find structure in the data.

Clustering

Group similar datapoints together.

Anomaly detection

Finds unusual data points, e.g. a faulty sensor reading.

Dimensionality reduction

Compress data into fewer variables while keeping its structure.

DeepLearning.AI. Phoon, K. K., Ching, J. & Cao, Z. (2022). Unpacking data-centric geotechnics. *Underground Space* 7(6), 967–989. Chart data: UCI Concrete Compressive Strength (Yeh, 1998).

A SIMPLE SUPERVISED-LEARNING PICTURE · IT FOLLOWS US ALL COURSE

X is what we give the model. y is what we want it to predict.

1 Training

Cement	Slag	Fly ash	Water	SP	Coarse	Fine	Age	Strength
540.0	0.0	0.0	162	2.5	1040	676	28	79.99
332.5	142.5	0.0	228	0.0	932	594	270	40.27
198.6	132.4	0.0	192	0.0	978	826	360	44.30
… 1,030 mixes · inputs (X) in kg/m³ and days · output (y) in MPa								

Provide this dataset to an ML model for training.

→ **ML algorithm**

learns a relationship f so that f(X) ≈ y

2 Prediction

Cement	Slag	Fly ash	Water	SP	Coarse	Fine	Age	Strength
350.0	0.0	0.0	180	0.0	1000	800	28	?

A new mix: we know X, but not y yet.

→ **Trained model f**

applies what it learned

→ $\hat{y}$ **Predicted compressive strength (MPa)**

Training rows from the UCI Concrete Compressive Strength dataset (Yeh, 1998). The prediction row is an illustrative new mix.

WORDS WE WILL USE ALL COURSE

A small vocabulary goes a long way

X Features

The inputs we give the model. Also called variables.

`cement, water, age …`

y Target

The response we want to predict. In a training table it is the label.

`compressive strength (MPa)`

$\hat{y}$ Prediction

The model's estimate of y for a new input.

`ŷ = f(X)`

f Model

The relationship learned from the examples.

`a curve, a tree, a network …`

Dataset

Rows are examples; columns are features and the target.

`1,030 rows × 9 columns`

Training

Fitting the model to examples where y is known.

`model.fit(X, y)`

THE SIGNATURE OF THIS COURSE

One ML workflow. Multiple civil engineering applications.

01

Engineering problem

What do we need to estimate?

02

Data

Measurements, simulations or records

03

Python

Inspect, prepare, analyse

04

ML model

Learn patterns from examples

05

Decision

Interpret with domain knowledge

ENGINEERING PROBLEM	TARGET	ML PROBLEM
Concrete	Strength	Regression
Buildings	Energy load	Regression
Steel inspection	Fault class	Classification
Structural health	Stability class	Classification

QUICK CHECK · TWO MINUTES

Regression, classification or unsupervised?

01

Predict the settlement of a shallow footing in millimetres

Regression Classification Unsupervised

02

Label a soil sample as clay, silt or sand from its lab tests

Regression Classification Unsupervised

03

Group traffic sensors that show similar daily patterns

Regression Classification Unsupervised

04

Flag strain-gauge readings on a bridge that look unusual

Regression Classification Unsupervised

05

Estimate a building’s heating load from its geometry

Regression Classification Unsupervised

06

Decide whether a slope is stable or unstable

Regression Classification Unsupervised

QUICK CHECK · ANSWERS

The target decides, every time

01

Predict the settlement of a shallow footing in millimetres

Regression

A number on a continuous scale.

02

Label a soil sample as clay, silt or sand from its lab tests

Classification

Three possible categories.

03

Group traffic sensors that show similar daily patterns

Unsupervised · clustering

No labels; we look for groups.

04

Flag strain-gauge readings on a bridge that look unusual

Unsupervised · anomaly detection

We look for points unlike the rest.

05

Estimate a building’s heating load from its geometry

Regression

An energy quantity: a number.

06

Decide whether a slope is stable or unstable

Classification

Two possible categories.

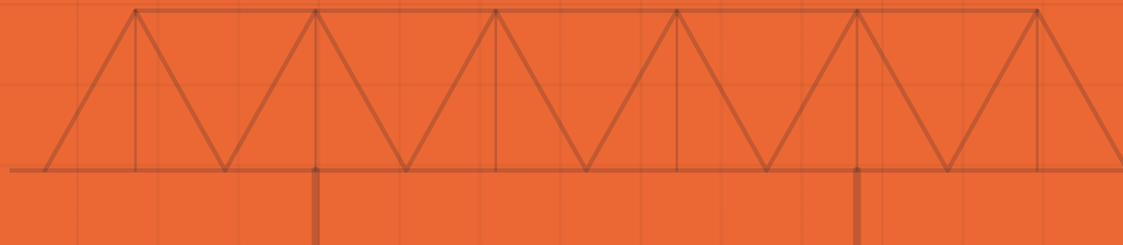
02

- a Why Python
- b Google Colab & notebooks
- c Colab, Anaconda and MATLAB
- d Our Python toolkit



PART 02

Python & Google Colab



PYTHON

Why are we using Python?

Open-source

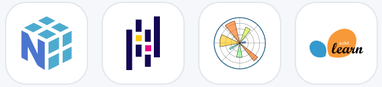
No paid language licence is required.

Easy to learn

An easy to learn programming language.

Scientific ecosystem

A huge ecosystem of libraries: NumPy, pandas, Matplotlib, scikit-learn and more.



Useful across disciplines

Data analysis, visualisation, ML and scientific computing.

OUR PRINCIPLE

We are not learning Python for its own sake.

We are learning enough Python to ask better engineering questions and build data-driven models.

WHAT IS IT WITH THE NAME? · WHY IS IT CALLED PYTHON?

Named after a comedy show, not a snake.

The name Python doesn’t come from the snake, even though the logo and community themes often use snake imagery.

Python was named after the British comedy group **“Monty Python’s Flying Circus”**, a favourite of Python’s creator, **Guido van Rossum**.

He wanted a name that was short, unique, and a little mysterious or fun, and “Python” felt just right.

FUN FACT



Python · Guido van Rossum

WHERE PYTHON FITS

Python supports the engineering workflow

01
Engineering problem
What do we need to estimate?

02
Data
Measurements, simulations or records

03
Python
Inspect, prepare, analyse

04
ML model
Learn patterns from examples

05
Decision
Interpret with domain knowledge

PYTHON SUPPORTS THE WORKFLOW

Inspect, prepare and analyse the data. Fit and evaluate the model.

- Load 1,030 test results in one line
- Plot strength against age
- Fit a model and score its error

ENGINEERING KNOWLEDGE SUPPLIES THE MEANING

Is this result physically plausible? Is it safe to act on?

- Is 80 MPa realistic for this mix?
- Are we extrapolating beyond the data?
- Is the error acceptable for design?

OUR ENVIRONMENT

What is Google Colab?

Hosted notebook environment
Write and run Python in a web browser.

Interactive
Combine code, text, equations and plots.

Low setup burden
No local Python installation is needed to begin.

Shareable
Notebooks can be shared and revisited online.

Google Colaboratory

FOR THIS WORKSHOP

Open the link → run a cell → inspect the output.

We can focus on engineering ideas rather than installation and environment setup.

Open: https://colab.research.google.com

ANATOMY OF A NOTEBOOK

What is a notebook?

Session01_first_steps.ipynb

File Edit View Insert Runtime Tools Help

Share

+ Code + Text

• Connected

TEXT CELL

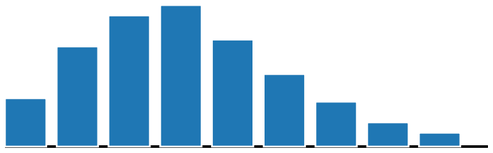
Water–cement ratio

Assumption: quantities in kg/m³. A lower w/c generally gives higher strength.

CODE CELL

```
import pandas as pd
df = pd.read_csv("concrete.csv")
df["w_c"] = df["water"] / df["cement"]  # new feature
df["w_c"].hist()
```

OUTPUT



Code cells

Run Python instructions and see the result beneath the cell.

Text / Markdown cells

Explain the engineering question, assumptions, equations and interpretation.

Outputs

Display numbers, tables, charts and model diagnostics in one place.

A notebook is a record of code, results and reasoning, not just a place to type code.

THE CHOICE WE MADE

Why Google Colab for this workshop?

No installation
Start in a browser.

Shared workflow
Everyone follows the same notebook sequence.

Easy to share
Links make distributing notebooks straightforward.

Live demonstrations
Run cells and inspect results together.

Engineering-focused
More time for data and interpretation.

Beginner-friendly
No environment management at the outset.

We choose Colab to reduce setup friction, not because it is best for every project.

ALTERNATIVES

Google Colab, Anaconda and MATLAB

FEATURE	Google Colab	Anaconda + Jupyter	MATLAB
Initial setup	**Browser; no local install to start**	Install and manage a local distribution	Installation and licence access may be required
Language	**Python**	Python (with R tools available)	MATLAB
Packages / tools	**Common libraries available; some setup may be needed**	Flexible packages and environments	MATLAB toolboxes and add-ons
Best fit here	**Shared, low-friction workshop**	Local work and controlled environments	Existing MATLAB workflows

All can support technical work; this workshop prioritises accessibility and shared hands-on learning.

GOOD TO KNOW

What should we know about Colab?

Internet access is needed
Colab runs in the browser.

Sessions are temporary
Save notebooks and important outputs.

Files and packages may need setup
Upload data or mount cloud storage when required.

Compute resources can vary
Hardware and usage limits are not guaranteed.

Protect sensitive data
Follow institutional data-protection rules before uploading data.

Use Colab as a practical workshop environment, not as a promise of unlimited compute.

TOOLS, INTRODUCED WHEN WE NEED THEM

Our Python toolkit

Python	Language and logic	`w_c = water / cement`
NumPy	Numerical arrays and calculations	`np.mean(strength)`
pandas	Tables and data preparation	`pd.read_csv("concrete.csv")`
Matplotlib	Plots and visual diagnostics	`plt.scatter(df.age, df.strength)`
scikit-learn	Machine-learning models and evaluation	`model.fit(X, y)`

We introduce tools when an engineering question calls for them.

The goal is not to memorise every function. It is to understand **what each step does** and **whether its result makes engineering sense.**

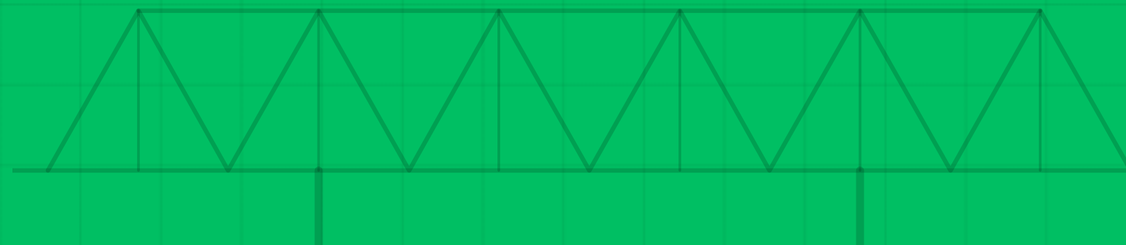
03

- a Open and run the notebook
- b Our first cell
- c Read the output like an engineer
- d Working with notebooks

PART 03 · HANDS-ON

Let’s get started with Python in Google Colab.

Dr. Abdul Waris · Research Associate, Universität Siegen



HANDS-ON · STEP BY STEP

How to open and run the workshop notebook



STEP 1

Open the shared Colab link

`https://colab.research.google.com`

STEP 2

Sign in to Google if prompted

Any Google account works.

STEP 3

Open the provided notebook

An .ipynb file, one per session.

STEP 4

Run a cell

using ▶ or `Shift` + `Enter`

STEP 5

Read the output before moving on

Does it make engineering sense?

⚠ If a cell fails, read the error and check the preceding steps before rerunning everything.

AN ENGINEERING-FLAVOURED EXAMPLE

Let's run our first cell

Session01_first_steps.ipynb [1] · Python 3

```
cement = 350    # kg/m³
water  = 180    # kg/m³
water_cement_ratio = water / cement
print(water_cement_ratio)
```

EXPECTED OUTPUT

```
0.5142857…
```

Who does what?

The computer performs arithmetic; **the engineer interprets what the ratio means.**

Try it

Change the water content. Observe what changes, and what the calculation alone cannot tell you.

FROM ONE MIX TO 1,030 MIXES

Load the whole dataset with pandas

Session01_first_steps.ipynb [2]

```
import pandas as pd
df = pd.read_csv("concrete.csv")
df.shape
```

OUTPUT

```
(1030, 9)
```

pandas reads the whole lab record into a table we can inspect, filter and plot.

1,030 rows: one row is one mix tested in the lab. **9 columns:** eight inputs and the strength.

X 8 inputs *y* strength (MPa)

[3] `df.head()`

	cement	slag	ash	water	superplastic	coarseagg	fineagg	age	strength
0	540.0	0.0	0.0	162.0	2.5	1040.0	676.0	28	79.99
1	540.0	0.0	0.0	162.0	2.5	1055.0	676.0	28	61.89
2	332.5	142.5	0.0	228.0	0.0	932.0	594.0	270	40.27
3	332.5	142.5	0.0	228.0	0.0	932.0	594.0	365	41.05
4	198.6	132.4	0.0	192.0	0.0	978.4	825.5	360	44.30

Data: Concrete Compressive Strength, UCI Machine Learning Repository (Yeh, 1998). Column names follow a common CSV release of the dataset.

OUR FIRST PLOT

One line of Matplotlib, 1,030 tests on screen

Session01_first_steps.ipynb [4]

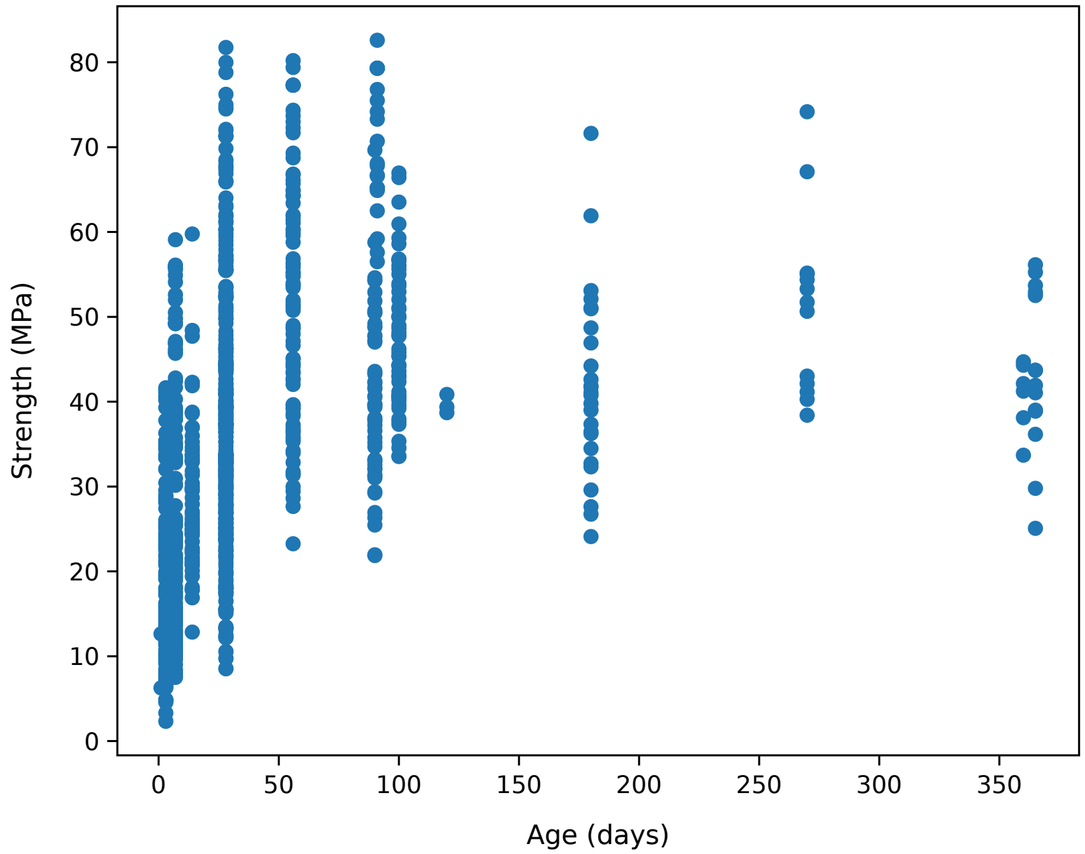
```
import matplotlib.pyplot as plt

plt.scatter(df["age"], df["strength"])
plt.xlabel("Age (days)")
plt.ylabel("Strength (MPa)")
plt.show()
```



What do you notice? Tests happen at fixed ages, strength rises with age, and the spread at every age is wide.

OUTPUT

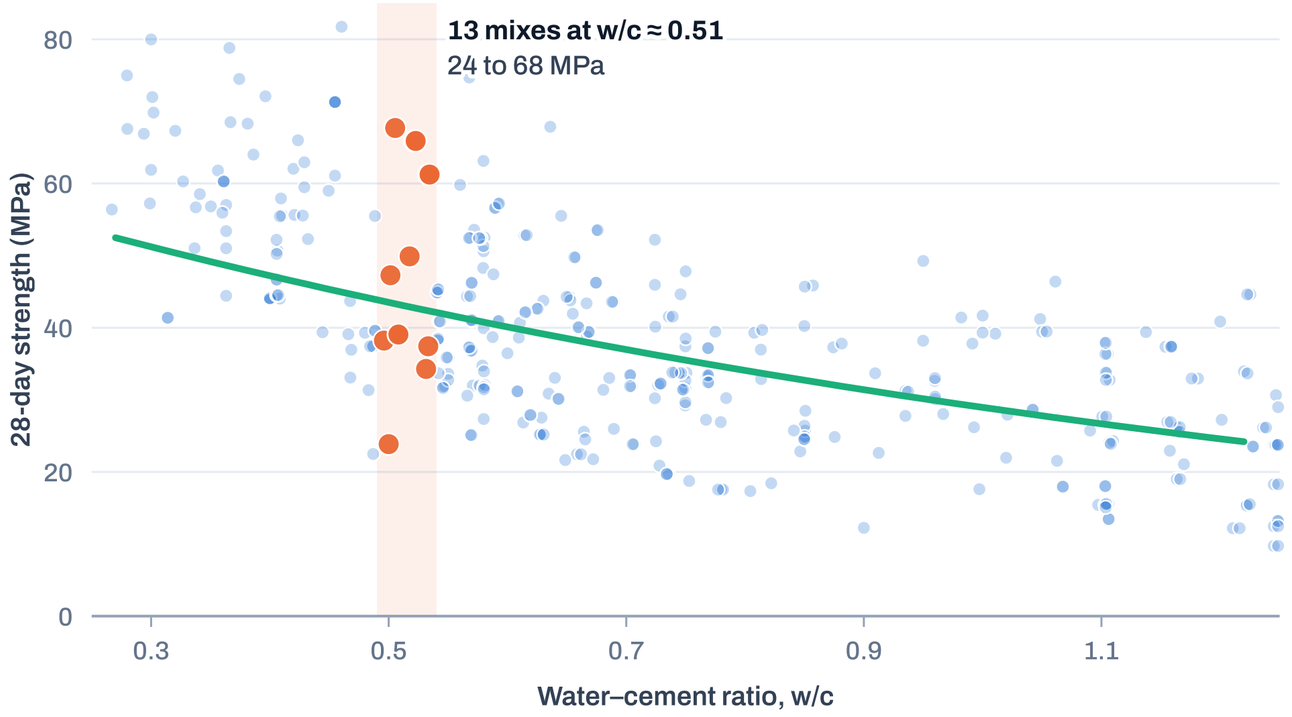


Data: UCI Concrete Compressive Strength (Yeh, 1998). The output is drawn in Matplotlib's default style.

READ THE OUTPUT LIKE AN ENGINEER

0.51 is a number. Is it a good mix?

Our w/c of 0.51 among 425 real mixes at 28 days




```
df["w_c"] = df["water"] / df["cement"]
```

24–68 MPa

13 mixes in the dataset share a w/c of about 0.51. Their 28-day strengths range from **24 to 68 MPa**.

WHAT THE CALCULATION CANNOT TELL YOU

Slag, fly ash, superplasticizer and aggregates matter too. Bringing all the inputs together is exactly what ML will help us do.

Data: UCI Concrete Compressive Strength (Yeh, 1998). Highlighted: 28-day mixes with 0.49 < w/c < 0.54. Curve: the Abrams-type fit from Part 01.

GOOD HABITS FROM DAY ONE

Working with workshop notebooks

01 **Open the shared link**
Use the notebook for the relevant session.

02 **Keep your own copy**
Use `File → Save a copy in Drive` when appropriate.

03 **Work in sequence**
Run cells from top to bottom unless instructed otherwise.

04 **Keep the analysis reproducible**
Record assumptions and note any changes to evaluation steps.

WORKSHOP NOTEBOOK LINK

Open today’s notebook and make your own copy.

Link shared in the session chat

SUMMARY

Why this setup?

Python

An accessible, widely used language for scientific computing and ML.

Google Colab

A browser-based notebook environment with minimal setup.

Engineering judgement

The goal is meaningful analysis, not merely code that runs.

We chose Colab to reduce installation barriers and keep the cohort focused on real Civil Engineering problems.

NEXT

Load a dataset, inspect its structure, and ask our first engineering questions.

BEFORE SESSION 02

Three small things to try

1

Run it all

Open today’s notebook, save your own copy, and run every cell from top to bottom.

```
File → Save a copy in Drive
```

2

Change the water

Try 160 and 200 kg/m³. Note the new w/c, and what it still cannot tell you.

```
water = 160
```

3

Bring a question

Write down one quantity from your own field you would like to predict, and the inputs you would need.

```
y = ? · X = ?
```

02 **From Raw Engineering Data to an ML-Ready Dataset**

Explore, clean, visualise and split engineering data.

Questions?



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END OF SESSION 01

Thank you.

See you in session 02: From Raw Engineering Data to an ML-Ready Dataset.

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